CCPROG2 (for S24B) MP Specifications – Part 2 of 3
User-Defined Functions for Taylor Series Expansion of Sine and Cosine
Part 2 is 40% of the MP Grade

I. INTRODUCTION

The 2nd part of the machine problem is concerned with an **improved** (i.e., better) implementation of the cosine and sine approximations by user-defined functions in C.

The objectives of Part 2 of the machine problem are:

1. to apply what you learned on the "divide and conquer method" with functions, and
2. to realize and (hopefully) appreciate the fact that functions provide a better solution to the Taylor Series expansion problem (compared with your previous implementation where all computations are inside the **`main()`** function), and
3. to apply and appreciate the concept of separate compilation, and linking

II. REQUIREMENTS

We will **restructure** the solution to the Taylor series expansion as a collection of **user-defined functions** instead of implementing everything inside the **`main()`** function.

You will be provided three source files for this activity, namely:

a. **`mp2_math.h`** – contains the **`#define`** for **`PI`** and four function prototypes
b. **`lastname_mp2_math.c`** – contains the skeleton of the function definitions corresponding to the function prototypes in **`mp2_math.h`**, and
c. **`lastname_mp2_main.c`** – contains the skeleton of the main() function

Your task is to implement the body of the function definitions in the skeleton programs (items b and c in the list above).

The output of your executable file should be the same and follow the same format as MP Part 1.

III. DELIVERABLES AND SUBMISSION DEADLINES

The policy on late submission is the same as in MP Part 1. You are strongly encouraged to submit on time.

1. Submit the following via MP2 Canvas submission page:
   a. the two C source files via the Canvas submission page; submission via email will NOT be accepted as Canvas is the official repository of all student submissions (not my email inbox).
   b. accomplished TEST Script document as a PDF file; rename the accompanying `LASTNAME-TestScript.docx` file using your last name, for example, if your last name is SANTOS then your file should be named as `SANTOS-TestScript.PDF`.

2. Submit an accomplished MP submission checklist (print and accomplish the accompanying document named "mp2_checklist.pdf") within the **first 10 minutes of our in-person class nearest the date after the MP2 Canvas submission deadline.**

IV. TESTING, CHECKING and GRADING SCHEME

The testing, checking and grading scheme is the same as in Part 1. In case your program turns out NOT to be 100% semantically correct, there is still one more chance (the last!) to fix it in Part 3 (final) part of this machine problem.

***** End of the Machine Problem (Part 2) Specifications *****

Last Note: Consult me immediately if you have any clarification, question or problem regarding this academic activity. Enjoy! ☺☺☺